



UNITED INTERNATIONAL UNIVERSITY

Department of Computer Science and Engineering

CSE 3522: Database Management Systems Laboratory

UIU Social

Submitted by:

Team 1

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1. Problem Statement

Academic, extracurricular, and everyday administrative operations at United International University (UIU) rely on a complex network of daily interactions between students, faculty members, club executives, and administrative staff. However, UIU currently lacks a unified, centralized digital ecosystem to support these critical workflows. Information and campus utilities are severely fragmented across disparate, non-institutional channels including informal Facebook groups, Discord servers, scattered emails, and physical notice boards. As a result, course announcements get buried under social clutter, official event updates fail to reach the broader student body, and peer-to-peer academic collaboration remains disorganized.

Furthermore, physical and media engagement gaps persist on campus. Newly enrolled students, faculty, and visitors frequently struggle to navigate UIU’s multi-story campus structure and locate specific faculty cubicles, laboratories, or seminar rooms due to the absence of an interactive digital floor directory. Additionally, while student clubs, faculty members, and tech groups regularly produce insightful discussions, research spotlights, and workshop recordings, there is no dedicated campus media channel to stream audio and video podcast content, forcing creators to rely on external platforms where university-specific reach is lost.

The individuals who face these problems most directly are students, course instructors, student-led club managers, and university administrators. Students constantly struggle to keep track of section-specific assignments, locate unfamiliar campus rooms, and catch up on recorded campus tech talks dispersed across unofficial groups. Faculty members lack a dedicated digital space for informal academic interaction, room navigation assistance for guest lecturers, or research assistant recruitment beyond formal grade submission portals. Club executives suffer from low event attendance and limited media reach because there is no official, central hub to publicize workshops, stream podcast highlights, or manage RSVPs. System administrators also lack moderation control over university-related public groups, exposing the community to unverified accounts and potential misinformation.

This structural fragmentation creates operational inefficiency, security risks, spatial navigation hurdles, and communication delays. Important academic notices and exam schedules are frequently missed, while informal communication platforms lack mandatory identity verification using official institutional emails (@uiu.ac.bd). If this issue is not addressed, academic engagement will continue to suffer from communication barriers, extracurricular club participation will decline, campus navigation will remain cumbersome, and direct student-to-faculty academic interaction will remain restricted.

The main gap in the current ecosystem is not the absence of communication tools; it is the absence of a verified, academic-first central database that connects students, faculty, and clubs within a structured institutional boundary. **UIU Social** is proposed to bridge this gap by establishing a secure, role-restricted relational database system that unifies section-based study groups, instant peer-to-peer messaging, verified user profiles, campus event management, interactive 2D floor maps, an audio/video podcast hub, and club activity hubs into a single platform.

2. Motivation

We selected this project because academic collaboration and campus community building are central to university life, yet day-to-day communication at UIU remains dependent

on third-party commercial platforms. Building a dedicated university network provides a practical, real-world domain to solve authentic campus problems while applying advanced database principles such as relational entity design, role-based access security, relational joins, triggers, and full-text search indexing.

A centralized campus platform directly benefits every stakeholder group. Students receive structured, section-specific course rooms and a clean feed for academic resources. Faculty members obtain a verified channel to post updates and highlight research opportunities. Student clubs gain an official platform to showcase activities and track event registrations. University administrators gain a secure moderation system to verify user identities and maintain academic decorum.

Existing solutions fall short of meeting these requirements. While existing learning management systems (such as UIU ELMS) handle formal grading and course material delivery, they lack instant messaging, interactive social feeds, student-driven posts, and club activity spaces. Commercial social networks like Facebook provide dynamic feeds but lack academic structure, institutional verification, and section-level privacy. **UIU Social** takes a targeted approach: it does not seek to replace formal LMS gradebooks, but rather bridges the gap between structured academic management and dynamic campus social communication through a secure, relational database framework.

3. Objectives

1. To implement a secure authentication system requiring institutional email validation (@uiu.ac.bd) and role-based access control for Students, Faculty, and Administrators.
2. To establish an administrative verification queue and moderation dashboard to validate user credentials and maintain content safety.
3. To construct section-specific academic course groups and department hubs for resource sharing and discussion.
4. To provide a centralized Campus Events and Seminars Hub supporting real-time event discovery, category filtering, and RSVP tracking.
5. To build an interactive Club Hub showcasing campus organizations, executive boards, and event announcements.
6. To enable direct 1-on-1 real-time messaging between verified users with secure file attachment processing.
7. To develop customizable academic user profiles featuring role badges, research links, and academic interest tags.
8. To maintain an interactive central community feed for official university announcements and peer-to-peer updates.
9. To design a global search engine indexing users, course sections, campus clubs, and events.
10. To enforce data integrity, relational consistency, and automated notification triggers across all user interactions.

4. Competitor Analysis

We analyzed three primary systems currently used for academic and social communication within university communities. While each system addresses specific operational needs, none offer an integrated, verified solution tailored to UIU’s campus ecosystem.

4.1 Facebook Groups & Commercial Social Media

Facebook Groups offer high user familiarity, dynamic newsfeeds, and media sharing. However, they lack official university verification, operating without mandatory institutional email authentication. Academic discussions easily get buried under informal social content, and groups lack section-specific security or institutional role badges.

4.2 Discord Workspaces

Discord provides well-organized textual and voice channels, customizable role permissions, and smooth file exchange. However, Discord lacks structured academic profile systems, integrated event discovery calendars, interactive club hubs, and automated verification against official university student rosters.

4.3 UIU ELMS (eLearning Management System)

UIU ELMS acts as the official academic repository for grade distribution, assignment submission, and course slides controlled by instructors. However, ELMS is completely rigid and lacks peer-to-peer messaging, student-led posting feeds, extracurricular club spaces, and interactive event RSVP tracking.

System	Main Strength	Gap Relative to Our Scope	UIU Social Focus
Facebook Groups	High user familiarity, dynamic media feeds, easy event creation.	No institutional email validation; high clutter; lacks academic structure.	Mandatory @uiu.ac.bd verification, role badges, and structured course groups.
Discord Workspaces	Excellent real-time chat organization, channel roles, file sharing.	Lacks built-in event discovery, club showcases, and academic profiles.	Integrated event calendar, club hub, and verified student/faculty profiles.
UIU ELMS	Formal course delivery, assignment submissions, grade management.	Lacks student posting, peer messaging, club hubs, and instant chat.	Dynamic social feeds, real-time 1-on-1 messaging, and club spaces alongside courses.
Google Maps	Outdoor geographical navigation and landmark pinpoints.	Lacks indoor building layouts, multi-floor room listings, and room/faculty tags.	Interactive 2D floor-by-floor maps for classroom, lab, and faculty office location.
Social Media	Global video host and media streaming capabilities.	No campus-specific community filtering or direct link to academic course groups.	Centralized audio and video podcast hub for campus tech talks and student shows.

5. Features

5.1 Account Verification & Role-Based Authentication

What it does: Provides signup requiring institutional email (@uiu.ac.bd), student/faculty ID, and department details. Submissions undergo a mandatory verification process against institutional records before access is granted.

Who will use it: Students, Faculty Members, and System Administrators.

Why it is needed: Guarantees that only legitimate university members gain access to internal academic channels and private group communications.

5.2 Admin Verification Queue & Moderation Dashboard

What it does: Gives administrators a centralized panel to review pending user registrations, approve or reject applications, manage reported content, and review safety flags.

Who will use it: System Administrators.

Why it is needed: Maintains platform safety, prevents identity spoofing, and enforces institutional codes of conduct.

5.3 Academic Department & Course Groups

What it does: Automatically routes verified users into department channels (e.g., CSE, EEE) and course-section rooms (e.g., DBMS Sec B). Features include resource sharing, group joining/leaving, and announcement feeds.

Who will use it: Students and Course Instructors.

Why it is needed: Keeps academic discussions localized and structured, preventing general feeds from becoming cluttered.

5.4 Community Social Feed & Announcement Board

What it does: Delivers an interactive central feed for academic updates, questions, image sharing, and official faculty notices (e.g., exam schedules, library hours).

Who will use it: All platform members.

Why it is needed: Guarantees high visibility for campus-wide announcements and fosters peer-to-peer academic discussions.

5.5 Campus Events & Seminars Hub

What it does: Centralized portal showcasing upcoming seminars, hackathons, and workshops with filters for dates and categories. Includes live attendee counts and an "RSVP / Secure Spot" button.

Who will use it: Students, Event Organizers, and Club Executives.

Why it is needed: Streamlines event publicity and simplifies seat booking for campus activities.

5.6 Interactive Club Hub & Categorization

What it does: Dedicated showcase space for official university clubs (e.g., Robotics Club, Computer Club) featuring category filters (Tech, Cultural, Sports), executive team lists, and activity feeds.

Who will use it: Club Executives and General Students.

Why it is needed: Increases student participation in extracurricular activities by giving clubs an official voice.

5.7 Direct Real-Time Messaging & File Exchange

What it does: Enables direct 1-on-1 messaging between users with online status indicators, document attachment capability, and notification controls.

Who will use it: Students and Faculty Members.

Why it is needed: Allows convenient private communication for project guidance, research, and peer collaboration.

5.8 Automated Notifications & Reminders

What it does: Triggers automatic alerts for unread direct messages, new course group announcements, and upcoming RSVPed events.

Who will use it: All verified platform members.

Why it is needed: Ensures users never miss critical academic deadlines or registered events.

5.9 Academic User Profiles

What it does: Displays customizable user profiles featuring official role badges (e.g., Assistant Professor, Student), research publication links, academic interests, and past post histories.

Who will use it: Faculty Members and Students.

Why it is needed: Helps students showcase academic achievements and assists faculty members in identifying student research assistants.

5.10 Global Search & Filtering Engine

What it does: Site-wide search box allowing users to query events, course groups, campus clubs, faculty members, and student directories.

Who will use it: All platform members.

Why it is needed: Provides fast access to platform resources without manual menu navigation.

5.11 Interactive 2D Campus Floor Map

What it does: Displays floor-by-floor 2D interactive layouts of university buildings, allowing users to search and locate classrooms, labs, faculty rooms, and campus amenities.

Who will use it: Students, Faculty Members, and Campus Visitors.

Why it is needed: Helps users navigate the campus easily and reduces time spent searching for unfamiliar room locations.

5.12 Audio & Video Podcast Hub

What it does: A dedicated media portal hosting recorded and live-streamed audio and video podcasts covering technical talks, club shows, and academic interviews.

Who will use it: All verified platform members.

Why it is needed: Provides a central channel for university-wide video/audio content.

6. Technology Stack

Layer / Tool	Planned Technology	Reason for Selection
Frontend	HTML5, CSS3, JavaScript (ES6)	Builds responsive user interfaces, dynamic feeds, and messaging panels without adding heavy framework dependencies.
Backend	PHP	Efficient server-side execution for session authorization, HTTP request routing, and direct database queries.
Database	MySQL	Relational database system offering foreign keys, ACID compliance, joins, and indexing for user and group relationships.
Environment	XAMPP / Apache	Simplifies local Apache server and MySQL database testing during development.
Development Tools	VS Code, Git, GitHub	Provides robust version control, collaborative code management, and issue tracking.
UI Design	Figma	Allows the team to prototype page layouts and user interaction flows before coding.

7. Feature Distribution Among Team Members

The system features and backend development responsibilities have been strictly balanced across all team members based on technical complexity and execution workload.

7.1 1. Md. Kawsar Ahmed (ID: 0112430083) - Core Messaging & Security Lead

Assigned Features:

1. Direct Real-Time 1-on-1 Messaging, Chat Histories & Secure File,Image,Document Attachments
2. Institutional Email Validation (@uiu.ac.bd)
3. Role-Based Access Control (RBAC) & Admin Verification Queue
4. Content Moderation Dashboard & Flagged Content Purger
5. 2D Map Integration for All Floors & Audio,Video Podcast Section

Main Responsibilities: Building AJAX-based real-time chat communication, enforcing file MIME-type security, writing backend security scripts, managing admin approval queues, and handling account moderation.

7.2 2. Avijit Saha (ID: 0112430535) - Team & Technical Lead

Assigned Features:

1. Global Search & Multi-Table Relational Indexing Engine
2. Academic User Profiles & Student Portfolio Showcase Module
3. System-Wide Responsive Shell Layout & UI Master Templates
4. Full-Text Query Optimization & Cross-Module Data Integration

Main Responsibilities: Architecting primary relational tables, optimizing multi-table search query performance, designing master UI templates, and overseeing end-to-end project integration.

7.3 3. Shaikh Mahmudul Hasan Emon (ID: 0112430105)

Assigned Features:

1. Department Rooms & Course Section Directories (e.g., DBMS Sec B)
2. Central Interactive Community Feed Layout
3. Official Announcement Board for Faculty Updates
4. Academic Feed Post Interaction Module (Edit, Pin, Purge)

Main Responsibilities: Constructing SQL join logic for department and course section rosters, designing central feed templates, and managing course group interaction routines.

7.4 4. Tamim Al Mitul (ID: 0112430115)

Assigned Features:

1. Campus Events & Seminars Hub Portal
2. Live Event Participant Registration & RSVP Counter Engine
3. Interactive Club Directory Hub & Categorization
4. Club Executive Board Management & Event Notices

Main Responsibilities: Designing relational tables for events and clubs, writing SQL queries for RSVP transactions, and building client-side filter scripts.

7.5 5. Molla Nabil Basar (ID: 0112430523)

Assigned Features:

1. Event Reminder
2. Central Unread Notification Alert Center
3. User Activity Logging & Event Update Sync
4. System Alert History & Preference Management

Main Responsibilities: Developing backend notification triggers, managing automated class/event reminder schedules, and configuring activity database logs.

8. Additional Sections

8.1 Target Users

- **Students:** For section-specific course discussions, event RSVPs, messaging, and club participation.
- **Faculty Members:** For posting course updates, sharing research opportunities, and communicating with students.
- **Club Executives:** For promoting campus events, sharing announcements, and recruiting new members.
- **System Administrators:** For user verification, content moderation, and platform security.

8.2 Proposed System Overview

UIU Social is a browser-based application built on a MySQL relational database. The frontend delivers role-specific user dashboards, while the backend applies strict security rules. The initial version focuses on mandatory identity verification, section-based study groups, campus events, club hubs, direct messaging, user portfolios, and administrative moderation queues. Following course guidelines, this report omits ERDs and schema designs.

8.3 Scope of the First Version

- Restricted exclusively to verified UIU community members (@uiu.ac.bd).
- Web-based user access through responsive desktop browser layouts.
- Relational database tables for user roles, course sections, event RSVPs, and messages.
- Role-restricted access across Students, Faculty, Club Executives, and Admins.

8.4 Basic Use Cases

1. A student registers using an institutional email and waits for administrative verification.
2. An administrator reviews pending applications and approves verified profiles.
3. A faculty member posts an assignment update within a specific course-section room.
4. A club executive creates an event page for a seminar and tracks student RSVPs.
5. A student messages a course instructor to ask a question and attaches a document.
6. A student uses the global search bar to locate course groups, events, or faculty profiles.

8.5 Expected Outcomes

The project will deliver a fully functional, database-backed web application demonstrating central campus communication. The database will enforce data integrity, prevent unverified access, update event totals accurately, and maintain clear separation between course sections.

8.6 Limitations

- User registration relies on manual identity review against university rosters.
- Financial transactions for paid event tickets or membership fees are out of scope.
- The first release focuses on desktop web browsers rather than native mobile applications.

8.7 Future Improvements

- Automated identity verification using direct university registrar API integration.
- Native mobile applications for iOS and Android with push notifications.
- Integration with UIU ELMS for direct grade updates and course roster synchronization.

8.8 Four-Week Development Timeline

Week	Main Activities	Expected Output
Week 1	• Kawsar: Institutional email validation, RBAC database tables, attachment schema. • Avijit: ERD layout, master UI templates, system-wide responsive shell layout. • Emon: Department rooms & course section directory table architecture. • Tamim: Campus Events & Interactive Club Directory relational tables. • Nabil: Central Unread Notification tables & user activity logging schema.	Approved ERD, SQL DDL script execution, responsive shell layout, and email validation setup.
Week 2	• Kawsar: Direct real-time 1-on-1 messaging backend, chat history, audio/video podcast section. • Avijit: Academic user profiles, student portfolio showcase, global search query foundation. • Emon: Central Interactive Community Feed & official faculty announcement board. • Tamim: Interactive club directory categorization & club executive board management. • Nabil: System alert history, preference management & user activity logging middleware.	Functional messaging system, working user profiles, interactive feed, and logging framework.
Week 3	• Kawsar: Content moderation dashboard, admin verification queue, 2D floor map integration. • Emon: Academic feed post interaction module (Edit, Pin, Purge) & course section rosters. • Tamim: Live event participant registration & RSVP counter engine using ACID transactions. • Nabil: Automated event reminders & event update sync background routines. • Avijit: Multi-table relational indexing engine & full-text query optimization.	Working RSVP transaction engine, 2D map navigation, admin panel, and event reminders.
Week 4	• Kawsar: Attachment MIME-type security audit, map/podcast query verification, moderation checks. • Avijit: End-to-end integration, full-text query optimization, query execution plans (EXPLAIN). • Emon: Cross-testing feed interactions, faculty announcement boards, and course section rooms. • Tamim: Concurrency testing for simultaneous event RSVPs & club permission checks. • Nabil: Notification Alert Center load testing & activity log cleanup routines.	Fully integrated DBMS application with index benchmarking, tested features, and final report.

9. Conclusion

UIU Social addresses campus communication challenges by consolidating fragmented interactions into a single, verified database application. By combining section-specific course rooms, real-time messaging, campus event discovery, club management, and administrative moderation into one platform, the system replaces informal social media groups with a secure academic workspace. The project serves as an ideal application of database management concepts, relying heavily on relational data models, role-based security, joins, triggers, and full-text indexing.

References

1. Meta Platforms, 'Facebook Groups Features & Management,' <https://www.facebook.com/community> (accessed August 2026).
2. Discord Inc., 'Discord Infrastructure & Permission Systems,' <https://discord.com/safety> (accessed August 2026).
3. United International University, 'eLearning Management System (ELMS),' <https://elms.uiu.ac.bd> (accessed August 2026).